

# Lakshay Chauhan

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## EDUCATION

**Indraprastha Institute of Information Technology (IIIT) Delhi**

Jul 2021 – Jun 2025

*Bachelor of Technology in Computer Science and Engineering*

*Delhi, India*

## WORK EXPERIENCE

**Member of Technical Staff**

Jul 2025 – Present

*ZL Technologies, Inc* | [Website](#)

*Hyderabad, India*

- Worked on task synchronization and bulk parallel processing across clusters within ZL's analytics module, resulting in over a 100% performance improvement.
- Implemented task completion time estimation and duplicate item detection when processing millions of items during data loading.

**Research Assistant** | [Gitlab Project Organization](#)

May 2024 – May 2025

*Networks and Systems Security Lab - IIITD* | [Website](#)

*Delhi, India*

- **Objective:** Address quality degradation in video streaming applications when fallback from QUIC to TCP occurs due to UDP blocking.
- **Solution:** Implemented a mechanism to send notifications in QUIC packets to anticipate UDP blocking. Server logs the kernel congestion window state to be used in the upcoming TCP connection. Improved user experience by avoiding TCP slow start, thus preventing buffering during fallback.

## PROJECTS

**Minimal Compiler Infrastructure** | [Repository](#)

May 2024 – Present

- **Key Skills:** *Compiler Infrastructure, Design and Optimization, Rust, FASM, WASM, Graphviz*
- Developed a minimal compiler infrastructure inspired by LLVM. It converts source code into an optimized control flow graph (CFG) and translates it into target-specific assembly code.
- Implemented CFG optimization passes, such as identifier validation, constant folding, and CFG simplification, ensuring high-performance code generation.

**Immediate-Mode GUI Framework** | [Repository](#)

Feb 2024 – May 2024

- **Key Skills:** *C/C++, Graphics Programming, Systems Programming, UI Frameworks, Rendering Pipelines*
- Designed and implemented a lightweight immediate-mode GUI framework from scratch using a low-level graphics library. Built a real-time rendering loop, input/event handling system, and modular UI components with minimal abstraction overhead.
- Focused on performance, explicit state management, and predictable memory usage for interactive applications.

**Needham Shcroeder Based PDF Print Server** | [Repository](#)

Mar 2025 – May 2025

- **Key Skills:** *C/C++, Networking, Cryptography, Client-Server Systems, Security Protocols*
- Implemented a secure client-server print service using the Needham-Schroeder authentication protocol. Designed custom request/response messaging and session validation to prevent unauthorized access. Gained hands-on experience with low-level networking, protocol correctness, and security-aware systems design.

## TECHNICAL SKILLS

**Languages:** C, C++, Rust, Assembly, Java, Kotlin, JavaScript, Python, Bash, Haskell

**Frameworks:** PyTorch, Django, ReactJS, Numpy, Pandas, Tauri, LibGDX, Raylib

**Developer Tools:** Emacs, Linux, Git, GDB, Markdown, Google Cloud Platform, OpenLiteSpeed, SQLite3

**Technical Electives:** Data Structures & Algorithms, Operating Systems, Cryptography, Database Management, Computer Security, Computer Networks, Compilers, Machine Learning, Natural Language Processing

## AWARDS

Awarded **Summer Undergraduate Research Fellowship** in 2023 by IRD-IIITD for the project “*Utilizing ultrasonic distance sensors as a mapping tool to design user-friendly CST*”.

Awarded **CHANAKYA Fellowship** by iHub Anubhuti Foundation in 2024 for the project “*A Unified Approach to User Emotion Detection through Emojis and Textual Analysis*”.